| TOOL OPTIONS | | | | |
|---|---|---|---|---|
| **Tool** | **Cost** | **Platform** | **Pros** | **Cons** |
| Ekahau HeatMapper | Free | Windows | Industry-grade accuracy | Windows-only |
| hnykda/wifi-heatmapper | Free | macOS/Linux/Win | Open source, CSV export | Manual calibration |
| Cisco Wireless 3D Analyzer | Paid | Web | Real-time 3D plus IoT sensors | Requires Cisco DNA license |

- **Routing table:** Core router handles inter-VLAN, switch runs Layer 2 only – simpler for troubleshooting.
- **Firewall zones:** Map VLANs to zones; block east-west traffic by default.

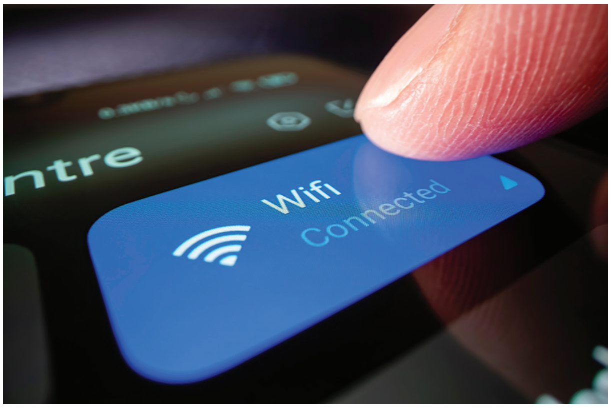
### Do this drill:

Use free software like Draw.io or LibreOffice Draw to create dual-layer diagrams: lock the physical layer, clone, and overlay logical icons – no re-drawing required when you move a VLAN.

## Heat-mapping your office Wi-Fi

### Why heat maps beat "Walk-Around Speed-Test"

RSSI-only speed tests miss dead zones hidden by glass cabins or steel cabinets. A genuine heat map reveals coverage holes before the CEO's call drops.




### Five-step survey procedure

1. **Upload floor plan:** High-res PNG in Ekahau or drag-drop into the GitHub tool.
2. **Scale drawing:** Mark two walls' real-world length so software can translate pixels to metres.
3. **Walk grid:** Hold the laptop at chest height; click each step on the map every 1-2 metres.
4. **Identify cold spots:** Look for anything below -70 dBm on 5 GHz, -67 dBm on 6 GHz.
5. **Reposition APs:** If an AP serves <15 m² of "green" (good) coverage, move it a metre or lower its power to cut overlap.

***Pro tip:*** *Don't crank transmit power past 18 dBm on Wi-Fi 6E radios – you'll just drown out upstream packets from client devices that can't shout back as loudly.*

**Validation pass:** After adjusting, re-run the survey. The "before" and "after" heat maps are your proof when finance asks why you need a third AP.

## Capacity planning: PoE budgets, UPS sizing & redundancy

### Calculate Your PoE budget

Modern Wi-Fi 6/6E APs draw up to 22.5 W (802.3at, aka PoE +). If eight such APs hang off a single 8-port PoE switch rated at 125 W, you're out of headroom – basic math: 22.5 W × 8 = 180 W > 125 W.

### Formula:

```
Total Power Draw = Σ(Device_Watts × Quantity)
Reserve Headroom  = Total_Power × 1.20
Switch Budget ≥ Reserve Headroom
```

- **Step 1:** List every PoE device – APs, IP phones, CCTV, IoT sensors.
- **Step 2:** Multiply by advertised max draw.
- **Step 3:** Add 20% headroom for cold-boot surges.

### Do this drill:

1. Open the PoE calculator at CablesAndKits or copy the above formula into Excel.
2. Plug in your device list; note any over-subscription in red.
3. If you're over budget, either split the load across two switches or upgrade to a higher-budget model.